

Demo: From Observed Treasury Yields to Hull–White Future Curves

A hands-on companion to Lecture 7

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

How can one observed curve support several future bond values without treating those scenarios as forecasts?

1 Goal

This demo follows one complete fixed-income workflow:

1. start with a **dated U.S. Treasury par-yield snapshot**;
2. bootstrap today's discount factors and zero-coupon yield curve;
3. calibrate Hull–White so that it exactly respects that curve;
4. fit the model's mean reversion and volatility to a supplied synthetic option quote snapshot that demonstrates the calibration mechanics;
5. generate yield curves at a future date in different interest-rate states; and
6. use those future curves to price remaining bond cash flows.

The initial curve uses the U.S. Treasury's published **January 2, 2025** par-yield snapshot. To keep every calculation auditable, the market nodes are embedded below rather than downloaded during rendering. Treasury par yields are bond-equivalent quoted rates; bootstrapped zero rates are continuously compounded. The option premiums used later are explicitly labeled synthetic because a stable, public option-volatility surface is not available with the course files.

Source: [U.S. Treasury, Daily Treasury Par Yield Curve Rates](#).



The central idea

Today's curve determines today's prices. Hull–White adds a disciplined way to describe *future* curves. Calibration makes those two statements consistent.

2 Setup

2.1 Read the workflow in two layers

Valuation today: observed par yields $\rightarrow$ teaching coupon bonds $\rightarrow$ discount factors $\rightarrow$ present value. **Uncertainty tomorrow:** supplied option premiums $\rightarrow$ fitted dynamics $\rightarrow$ future short-rate state $\rightarrow$ future curve $\rightarrow$ remaining bond value. The second layer must preserve the first. A complicated model is not allowed to change today's price of fixed cash flows once the discount curve is fixed.

Open **Show calculation** to inspect each step. The final state explorer changes the future state while holding the calibrated parameters and initial curve fixed.

We only need NumPy, pandas, and Matplotlib. The functions below use explicit inputs and return ordinary arrays or data frames so that each step is easy to inspect.

3 Step 1: Record the observed Treasury curve

The published Treasury series contains par yields at standard maturity nodes. We retain the 1-, 2-, 3-, 5-, 7-, and 10-year observations and linearly interpolate missing integer maturities. The interpolation is an **estimated curve-construction choice**, not an additional market quote. For transparent annual bootstrapping we treat each interpolated par yield as the coupon on an annual-pay par bond. A production Treasury curve would instead use security-level prices, semiannual coupons, actual day counts, accrued interest, and bills at the short end.

observation_date	maturity_years	observed_par_yield_pct
2025-01-02	1	4.17
2025-01-02	2	4.25
2025-01-02	3	4.29
2025-01-02	5	4.38
2025-01-02	7	4.47
2025-01-02	10	4.57

	maturity_years	par_yield_pct	im- plied_par_price
0	1	4.17	100
1	2	4.25	100
2	3	4.29	100
3	4	4.335	100

	maturity_years	par_yield_pct	im- plied_par_price
4	5	4.38	100
5	6	4.425	100
6	7	4.47	100
7	8	4.5033	100
8	9	4.5367	100
9	10	4.57	100

	input_status
0	Observed Treasury node
1	Observed Treasury node
2	Observed Treasury node
3	Interpolated node
4	Observed Treasury node
5	Interpolated node
6	Observed Treasury node
7	Interpolated node
8	Interpolated node
9	Observed Treasury node

Observed nodes are facts from the dated snapshot. Interpolated nodes and the annual-pay par-bond representation are assumptions. That separation matters: a curve can fit its inputs perfectly while still depending on construction choices.

4 Step 2: Bootstrap today's discount curve

For the one-year bond, the only cash flow is coupon plus principal, so its price immediately gives $P(0, 1)$. For each later bond, discount factors for earlier coupons are already known. The final discount factor is the only unknown:

$$P(0, n) = \frac{100 - \sum_{i=1}^{n-1} C_n P(0, i)}{100 + C_n}, \quad C_n = 100c_n.$$

	maturity_years	discount_factor	zero_rate_pct
0	1	0.96	4.0854
1	2	0.9201	4.1638

	maturity_years	discount_factor	zero_rate_pct
2	3	0.8815	4.2033
3	4	0.8437	4.2487
4	5	0.8068	4.2948
5	6	0.7707	4.3417
6	7	0.7355	4.3895
7	8	0.7019	4.425
8	9	0.6693	4.4613
9	10	0.6377	4.4984

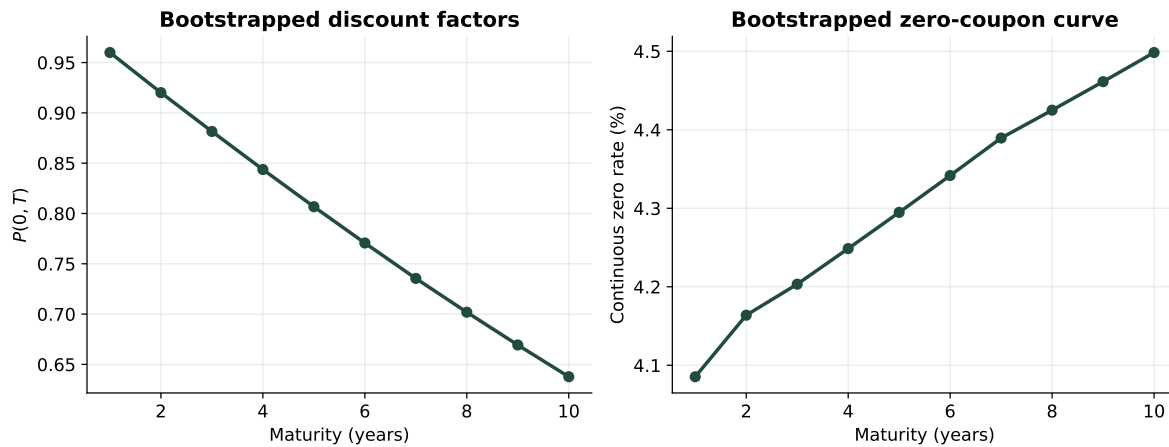


Figure 1: Today's curve recovered from observable coupon-bond prices

4.1 Check the bootstrap by repricing every input bond

A useful calibration habit is to return to the original observable quantities.

	maturity_years	par_yield_pct	im- plied_par_price
0	1	4.17	100
1	2	4.25	100
2	3	4.29	100
3	4	4.335	100
4	5	4.38	100
5	6	4.425	100
6	7	4.47	100
7	8	4.5033	100

	maturity_years	par_yield_pct	im- plied_par_price
8	9	4.5367	100
9	10	4.57	100

	input_status	repriced_from_curve	repricing_error
0	Observed Treasury node	100	0
1	Observed Treasury node	100	0
2	Observed Treasury node	100	0
3	Interpolated node	100	0
4	Observed Treasury node	100	0
5	Interpolated node	100	0
6	Observed Treasury node	100	0
7	Interpolated node	100	0
8	Interpolated node	100	0
9	Observed Treasury node	100	0

Largest absolute repricing error: 0.0000000000 per 100 face

5 Step 3: Make the initial curve usable at any maturity

i Bootstrap two maturities by hand

Under the annual-pay teaching convention, the 1-year discount factor is $P(0,1) = 100/104.17$. The 2-year coupon is 4.25, so $P(0,2) = [100 - 4.25P(0,1)]/104.25$. We remove the present value of the earlier coupon before solving for the final payment's discount factor. Using $1/(1 + 4.25\%)^2$ instead would confuse the par yield with a zero yield.

Hull–White formulas require $P(0, T)$ and the instantaneous forward rate $f(0, t)$. We use linear interpolation of the continuously compounded zero rate. This is intentionally simple; production systems use richer curve construction.

6 Step 4: Fit the Hull–White dynamics to option prices

The lecture’s calibration logic separates two jobs:

- the function $\theta(t)$ makes the model fit today’s *entire curve*; and
- a and σ control future mean reversion and volatility.

In practice, a and σ are fitted to liquid caps, floors, or swaptions. To keep the code transparent, our artificial option market consists of European calls on zero-coupon bonds. Such option premiums are observable, and Hull–White has a short closed-form pricing formula for them.

6.1 Hull–White building blocks

For the one-factor model

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

define

$$B(t, T) = \frac{1 - e^{-a(T-t)}}{a}.$$

The volatility of a zero-coupon bond option expiring at S on a bond maturing at T is

$$\sigma_P = \sigma B(S, T) \sqrt{\frac{1 - e^{-2aS}}{2a}}.$$

	option_expiry	bond_maturity	strike_per_100
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6.2 Create a synthetic option quote snapshot

We create at-the-money-forward calls with expiries from six months to five years. The hidden values $a = 0.18$ and $\sigma = 1.20\%$ are used only to generate the market premiums. A small alternating bid/ask-like perturbation prevents a mechanically perfect parameter recovery.

	option_expiry	bond_maturity	strike_per_100
0	0.5	5.5	80.4869
1	1	6	80.2801
2	1.5	6.5	80.1
3	2	7	79.9326
4	3	8	79.6205
5	4	9	79.3289
6	5	10	79.0493

	observed_option_price
0	0.8444
1	1.1124
2	1.2818
3	1.3817
4	1.5017
5	1.5362
6	1.5355

6.3 Calibrate a and σ with a simple grid search

The objective is the root-mean-square pricing error across the option quotes. A grid search is slower than a numerical optimizer, but students can see exactly what is tried and it needs no extra package.

```
Calibrated mean reversion a: 0.1790 per year
Calibrated volatility sigma: 1.1919% per sqrt(year)
Option-price RMSE: 0.005442 per 100 face
```

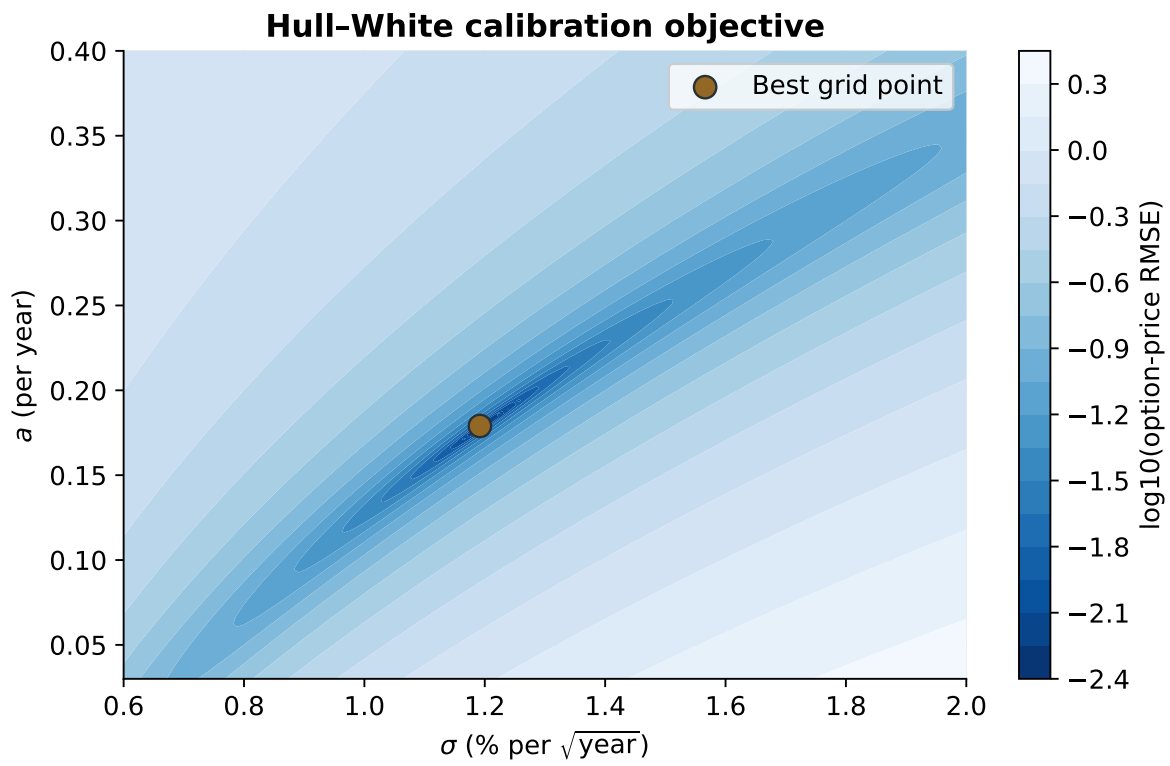


Figure 2: Option-pricing error across candidate Hull–White parameters

	option_expiry	bond_maturity	strike_per_100
0	0.5	5.5	80.4869
1	1	6	80.2801
2	1.5	6.5	80.1
3	2	7	79.9326
4	3	8	79.6205
5	4	9	79.3289
6	5	10	79.0493

	observed_op- tion_price	model_price	pricing_error
0	0.8444	0.8383	-0.0062
1	1.1124	1.11	-0.0023
2	1.2818	1.2743	-0.0075
3	1.3817	1.3807	-0.001
4	1.5017	1.4952	-0.0065
5	1.5362	1.5331	-0.0031
6	1.5355	1.5281	-0.0073

7 Step 5: Fit the time-dependent drift to today's curve

Once a and σ are known, the drift function that fits the observed initial curve is

$$\theta(t) = \frac{\partial f(0, t)}{\partial t} + af(0, t) + \frac{\sigma^2}{2a} (1 - e^{-2at}).$$

This is the lecture's key calibration point: $\theta(t)$ is a *function*, not one number. It supplies enough flexibility to match all maturities of today's curve.

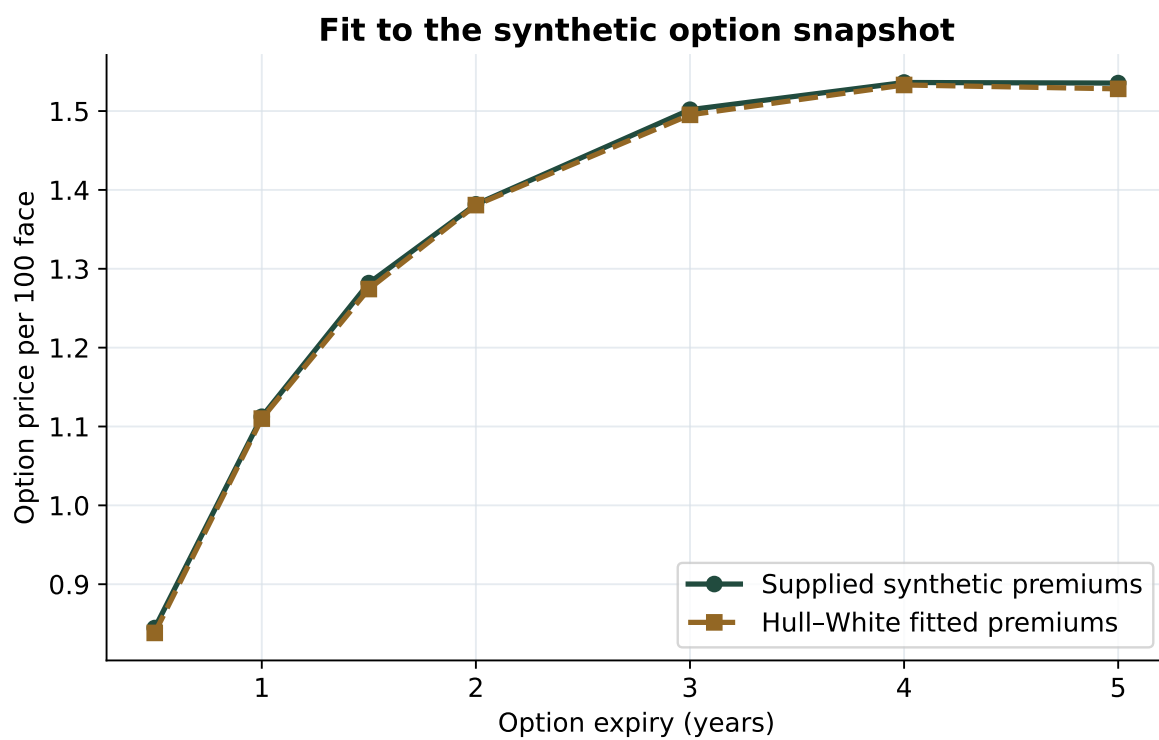
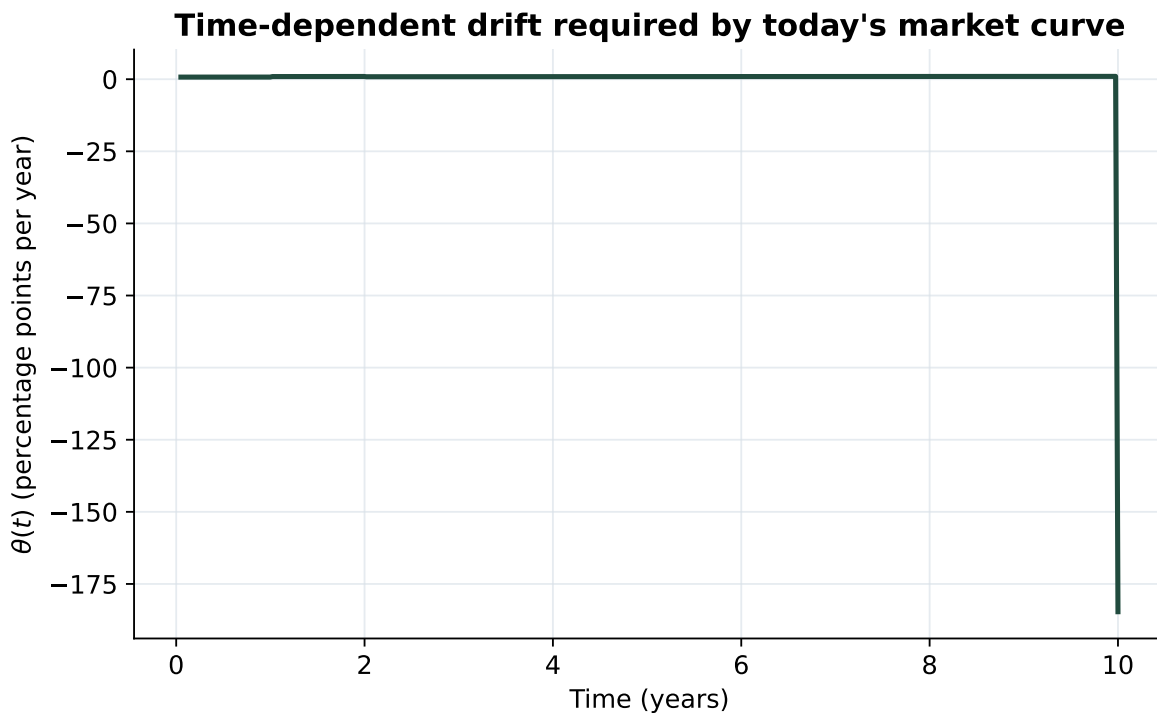


Figure 3: Observed artificial option premiums and calibrated Hull–White prices



The sharp changes are a consequence of using simple piecewise-linear interpolation. That is instructive: curve construction and smoothing matter in real calibration.

7.1 Verify that Hull–White reproduces today's discount curve

For future-date pricing, Hull–White gives the zero-coupon bond price

$$P(t, T) = A(t, T)e^{-B(t, T)r_t},$$

where

$$A(t, T) = \frac{P(0, T)}{P(0, t)} \exp \left[B(t, T)f(0, t) - \frac{\sigma^2}{4a}(1 - e^{-2at})B(t, T)^2 \right].$$

At $t = 0$, using $r_0 = f(0, 0)$ makes the exponential terms cancel, leaving exactly $P(0, T)$. This is the no-arbitrage fit.

	maturity	market_discount	hw_discount
0	1	0.96	0.96
1	2	0.9201	0.9201

	maturity	market_discount	hw_discount
2	3	0.8815	0.8815
3	4	0.8437	0.8437
4	5	0.8068	0.8068
5	6	0.7707	0.7707
6	7	0.7355	0.7355
7	8	0.7019	0.7019
8	9	0.6693	0.6693
9	10	0.6377	0.6377

	difference
0	0
1	0
2	-0
3	0
4	0
5	0
6	0
7	0
8	0
9	0

8 Step 6: Generate future yield curves

Move to year 2. In a one-factor Hull–White model, the short rate r_2 is the single state variable. Once a state is specified, `hw_discount_factor()` produces $P(2, T)$ for every $T > 2$, and therefore the entire future zero curve:

$$y(2, T) = -\frac{\log P(2, T)}{T - 2}.$$

We show three possible states around the model’s central short-rate level. They are scenarios, not point forecasts.

The center below is the initial instantaneous forward rate at year 2, chosen as a reference. It is not claimed to be the physical forecast or even the exact risk-neutral expected short rate; the latter also contains a volatility adjustment.

state	short_rate	maturity	zero_rate
Low-rate state	0.0278	2.25	0.0279
Low-rate state	0.0278	2.3481	0.0281
Low-rate state	0.0278	2.4462	0.0282
Central state	0.0428	2.25	0.0425
Central state	0.0428	2.3481	0.0426
Central state	0.0428	2.4462	0.0427
High-rate state	0.0578	2.25	0.0572
High-rate state	0.0578	2.3481	0.0571
High-rate state	0.0578	2.4462	0.0571

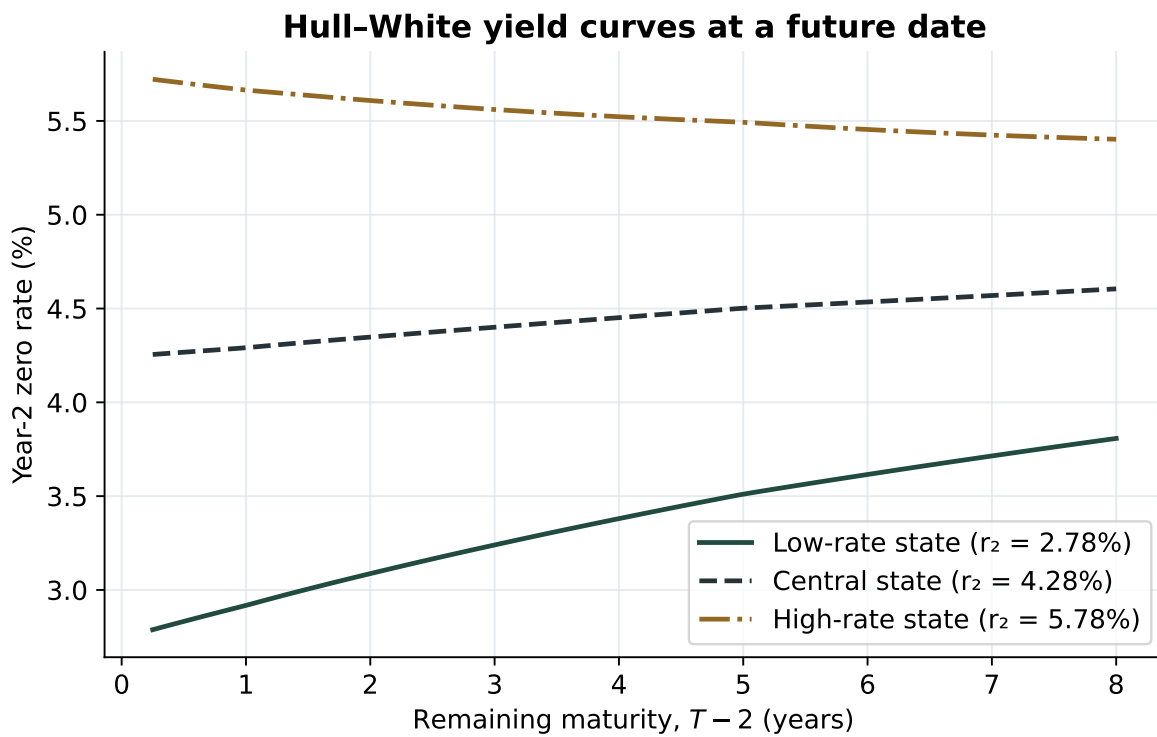


Figure 4: At year 2, one short-rate state determines a complete remaining yield curve

Notice that one factor moves the whole curve in a structured way. Mean reversion causes the effect of today's short-rate state to weaken at longer maturities because $B(t, T)$ approaches $1/a$ rather than increasing without bound.

9 Step 7: Use a future curve to price a bond

Consider the original ten-year 4.10% coupon bond at year 2, immediately after its year-2 coupon has been paid. Eight coupons and the principal remain. Its value in each state is simply the discounted value of those remaining cash flows using $P(2, T)$.

state	short_rate_pct	bond_value_at_year_2
Low-rate state	2.7817	101.8046
Central state	4.2817	96.106
High-rate state	5.7817	90.7425

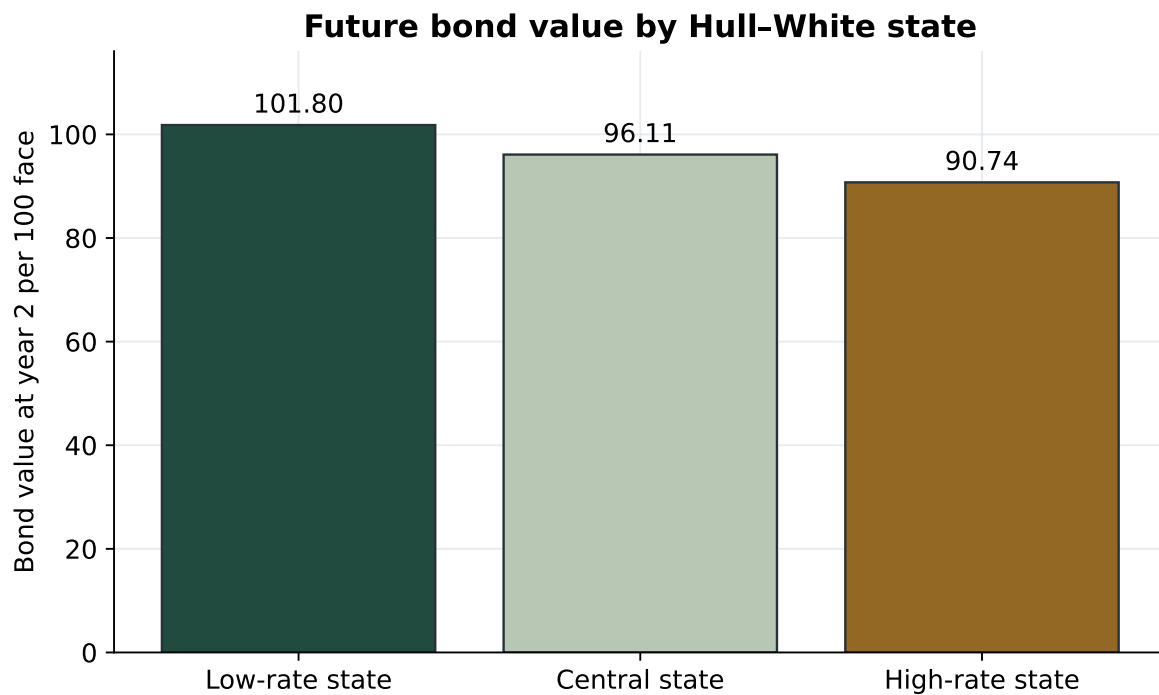


Figure 5: The same promised cash flows have different values in different year-2 rate states

This future valuation is where a short-rate model becomes useful. Today we do not observe $P(2, T)$; Hull-White generates it consistently for each possible r_2 .

10 Step 8: Plain-bond consistency at time 0

Finally, verify the lecture's important claim: once Hull–White is calibrated to the observed curve, it does **not** create a different time-0 price for a vanilla bond. Both methods discount the same fixed cash flows with the same $P(0, T)$.

```
Direct observed-curve price: 96.2742663687
Calibrated Hull-White price: 96.2742663687
Difference:                  0.000000000000
```

The equality is structural, not a coincidence. At time 0, the calibrated model uses the market discount factors by construction. For a vanilla bond, the model adds no new pricing information. Its value is the ability to generate future curves and to value cash flows that depend on those future states.

11 Checks and interpretation

The workflow has three separate fits, each with a different purpose:

Calibration object	Observable input	What is fitted	Why it matters
Today's discount curve	Coupon-bond prices	$P(0, T)$	Prices fixed cash flows today
Initial Hull–White term structure	Bootstrapped $P(0, T)$	$\theta(t)$	Makes the model no-arbitrage and curve-consistent
Future dynamics	Interest-rate option premiums	a, σ	Controls mean reversion and uncertainty
		passed	
Bootstrap reprices input bonds		True	
Discount factors are positive		True	
Discount factors decrease with maturity		True	
Hull–White matches today's discount curve		True	
Two vanilla-bond pricing methods agree		True	

12 Takeaways

Separate calibration from a scenario

First compare observed and interpolated nodes. Then inspect the option-fit error: an exact fit to today's curve does not guarantee an exact fit to every option. Finally change only the future short-rate state. Today's curve stays unchanged, but the future bond value changes. Explain why the year-2 value excludes coupons already paid and is not a two-year holding-period total return.

Market handoff: the embedded Treasury observations are a reproducible dated snapshot, not today's curve. Replace the entire dated node table together, retain the observed/interpolated labels, and rerun every check. Keep the supplied synthetic option prices if a permitted option dataset is unavailable; do not label recovered parameters as empirically calibrated market dynamics.

- Market bond prices can be converted into discount factors by bootstrapping.
- Hull–White does not fit a whole curve using only two constants. The function $\theta(t)$ fits today's term structure; a and σ describe future dynamics.
- Option prices provide information about those dynamics that vanilla bond prices do not.
- At a future date, each one-factor Hull–White short-rate state produces a complete yield curve and a value for the remaining cash flows.
- A vanilla bond has the same time-0 value under calibrated Hull–White and direct curve discounting. The model earns its keep when future rates affect valuation or cash flows.

Simplifications

Real desks use multiple instrument types, day-count conventions, accrued interest, bid/ask spreads, smooth curve interpolation, and robust numerical optimizers. The demo deliberately removes those layers so the economic logic stays visible.

13 Interactive explorer

Change the year-2 short rate relative to the reference forward rate. Keep today's Treasury curve and the fitted option parameters unchanged. The output is the ex-coupon value at year 2 of the remaining payments, not today's price or a holding-period return.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

14 Mini Project

Project brief. Use the fitted Hull–White model and the year-2 bond in this demo. Produce a memo with (1) the largest initial-curve repricing error, (2) the fitted mean reversion and volatility, (3) year-2 ex-coupon values for short-rate states 1 percentage point below and above the central state, and (4) the value of \$5 million face in each state. Explain why these are scenario values rather than expected physical returns.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

	Short rate (%)	Price per 100	Value of \$5mm face
0	3.2817	99.8666	4,993,328.512
1	4.2817	96.106	4,805,300.6097
2	5.2817	92.4944	4,624,719.0197

Mean reversion a	1.7899e-01
Volatility sigma	1.1919e-02
Largest initial-curve error	1.1102e-16
dtype: float64	

The initial curve pins down today's discounting; option premiums identify the model's risk-neutral dynamics. A higher year-2 short rate lowers the price of the same remaining payments. The \$5 million position scales the per-100 result by 50,000. These states are not physical forecasts or investment returns: the two earlier coupons and their reinvestment are absent from this ex-coupon valuation. Calibration here uses synthetic option quotes, so the fitted parameters demonstrate recovery of teaching inputs, not a live market fit.